Exploratory Analysis of TP53 and BRCA1 RNA-seq Counts in TCGA Breast Cancer Samples

Final Project Report

Shreya Ramesh

Foundations of Data Science

Dartmouth College

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1 Introduction

Breast cancer is a complex disease characterized by substantial molecular and clinical heterogeneity. Understanding variation in gene expression can provide insight into the biological processes associated with cancer development and progression. RNA sequencing (RNA-seq) is widely used to quantify transcript abundance and examine gene expression patterns across biological samples. Large-scale initiatives such as The Cancer Genome Atlas (TCGA) have generated molecular and clinical datasets that support exploratory analyses of cancer biology.

The TCGA Breast Invasive Carcinoma (TCGA-BRCA) project provides an opportunity to investigate RNA-seq measurements alongside clinical and sample metadata. The TCGA Network's comprehensive molecular characterization of breast tumors demonstrated the diversity of molecular alterations associated with breast cancer [1].

This project focuses on two biologically relevant genes: TP53 and BRCA1. TP53 encodes a tumor suppressor protein involved in cellular responses to DNA damage, cell-cycle regulation, and other processes that help maintain genomic stability. Alterations in the TP53 pathway are important features of many cancers, including breast cancer [1].

BRCA1 is another tumor suppressor gene with an important role in DNA damage repair, particularly homologous recombination. Disruption of BRCA1 function can compromise genomic stability and contribute to breast cancer susceptibility and tumor development [2].

The objective of this project is to explore the distribution and relationship of TP53 and BRCA1 RNA-seq counts in TCGA-BRCA samples. Specifically, the analysis examines descriptive statistics, the association between the two genes, differences in observed TP53 count distributions across sample types, and count patterns across ten cancer-related genes. The analysis also incorporates clinical metadata, including sample type and pathological stage, to provide context for the visualizations.

This study is exploratory and descriptive. Its purpose is to identify patterns in the available data rather than establish causal relationships or statistically significant differential gene expression.

2 Methods

2.1 Dataset and Data Preparation

This analysis used an RNA-seq count matrix and associated sample metadata from the TCGA-BRCA dataset. The count matrix contained 60,660 gene features across 1,231 samples, while the metadata file contained 1,231 observations and 98 variables.

The RNA-seq count matrix (`counts.csv`) and associated clinical metadata (`meta_data.csv`) were provided through the course materials and originate from The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) project. The dataset is associated with the National Cancer Institute's Genomic Data Commons (GDC), which provides access to TCGA molecular and clinical data [6]. The data was imported into R using the `read.csv()` function. Gene identifiers were retained as row names in the count matrix. The analysis selected TP53 (ENSG00000141510.18) and BRCA1 (ENSG00000012048.23) for descriptive statistics and comparative visualization.

To connect gene counts with sample information, the sample identifiers from the count matrix were standardized by replacing periods with hyphens. These identifiers were then matched to the corresponding barcodes in the metadata. Sample type was represented by the `definition` variable, while pathological stage was represented by `ajcc_pathologic_stage`.

2.2 Summary Statistics

Descriptive statistics were calculated for TP53 and BRCA1 across the 1,231 samples. These included the mean, median, standard deviation, minimum, and maximum RNA-seq count.

A separate frequency table summarized the distribution of samples across the three sample-type categories: primary solid tumor, solid tissue normal, and metastatic.

Pearson's correlation coefficient was calculated to describe the linear relationship between TP53 and BRCA1 raw counts.

2.3 Data Transformations and Visualizations

Several visualizations were generated to examine different aspects of the dataset.

A histogram was used to examine the overall distribution of TP53 counts. A scatter plot, with a fitted linear trend line, compared TP53 and BRCA1 counts across samples.

A box plot compared the medians, spread, and outliers of TP53 counts across sample types. An additional density plot examined the distribution of TP53 counts across these categories after applying the transformation

$$\log_2(\text{count} + 1).$$

This transformation reduced the visual influence of very large count values and allowed the sample-type distributions to be compared on a more compact scale.

A heatmap was also constructed using ten cancer-related genes: TP53, BRCA1, BRCA2, ERBB2, ESR1, EGFR, PTEN, MYC, KRAS, and BRAF. Counts were transformed using

$\log_2(\text{count} + 1)$ before visualization. Hierarchical clustering was applied to both genes and samples.

Two categorical annotations were included above the heatmap: sample type and pathological stage. These annotations provided additional clinical context for interpreting the observed count patterns.

2.4 Software and Packages

All analyses were conducted using R version 4.5.3. The `ggplot2` package, version 4.0.2, was used to generate the histogram, scatter plot, box plot, and density plot. The `ComplexHeatmap` package, version 2.26.1, was used to construct the annotated heatmap. The `knitr` package, version 1.51, supported the generation of the reproducible R Markdown report.

The visualizations were refined following peer feedback, with attention to figure dimensions, axis labels, legends, and overall readability. The use of `ggplot2` and `ComplexHeatmap` is consistent with their documented capabilities for statistical graphics and complex annotated heatmaps [4, 5].

3 Results

3.1 Sample Characteristics

The dataset contained 1,231 samples distributed across three sample-type categories. Primary solid tumor samples accounted for the majority of observations, followed by normal tissue and metastatic samples.

Table 1: Distribution of Samples by Sample Type

Sample Type	Number of Samples
Primary Solid Tumor	1111
Solid Tissue Normal	113
Metastatic	7
Total	1231

Table 2: Distribution of Samples by Pathological Stage

Pathological Stage	Number of Samples
Stage 0	4
Stage I	98
Stage IA	93
Stage IB	6
Stage II	7
Stage IIA	374
Stage IIB	267
Stage IIIA	155
Stage IIIB	27
Stage IIIC	62
Stage IV	20
Stage X	7
Missing	111
Total	1231

As shown in Table 1, the sample distribution was highly imbalanced. Primary solid tumor samples represented approximately 90.3% of the dataset, while normal tissue samples represented approximately 9.2%. Metastatic samples represented fewer than 1% of observations.

This imbalance is important when comparing distributions across sample categories, particularly for the metastatic group.

3.2 Summary Statistics for TP53 and BRCA1

Table 3 summarizes the observed RNA-seq counts for TP53 and BRCA1.

Table 3: Summary Statistics for TP53 and BRCA1 RNA-seq Counts

Gene	Mean	Median	SD	Min	Max
TP53	4935.74	4441	3036.71	274	25897
BRCA1	1304.21	919	1137.35	21	8050

TP53 had a mean count of 4,935.74 and a median of 4,441, compared with a mean of 1,304.21 and median of 919 for BRCA1. Both genes exhibited substantial variation across samples, as reflected by their standard deviations and count ranges.

For both genes, the mean exceeded the median, consistent with positively skewed count distributions. TP53 had a maximum observed count of 25,897, while BRCA1 had a maximum of 8,050.

Although TP53 had higher raw counts than BRCA1 in this dataset, this comparison does not establish that TP53 has higher normalized expression or greater biological activity. Raw counts can be influenced by sequencing depth, transcript length, and other technical factors.

3.3 Distribution of TP53 RNA-seq Counts

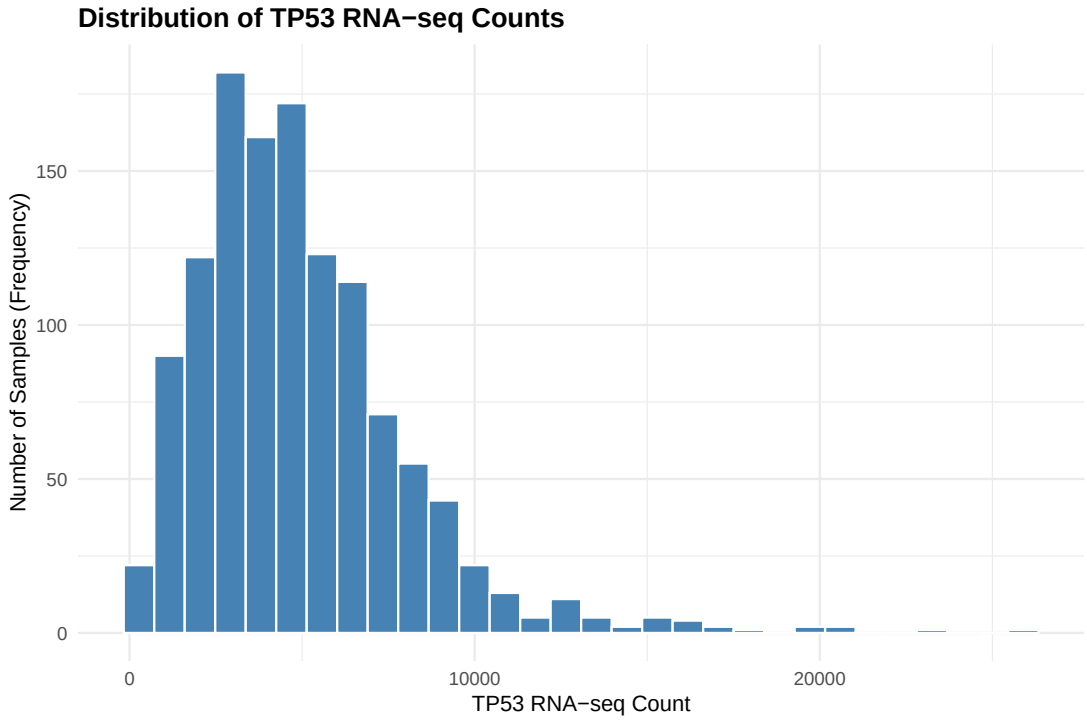


Figure 1: Histogram showing the distribution of TP53 RNA-seq counts across 1,231 TCGA-BRCA samples.

Figure 1 shows a right-skewed distribution of TP53 counts. Most samples were concentrated at lower to moderate count values, while relatively few samples had substantially higher counts.

The extended right tail indicates that a smaller subset of samples contributed particularly large observations. This pattern is consistent with the difference between the mean and median reported in Table 3.

The histogram demonstrates variability in observed TP53 counts and provides a useful starting point for exploring differences across individual samples.

3.4 Relationship Between TP53 and BRCA1

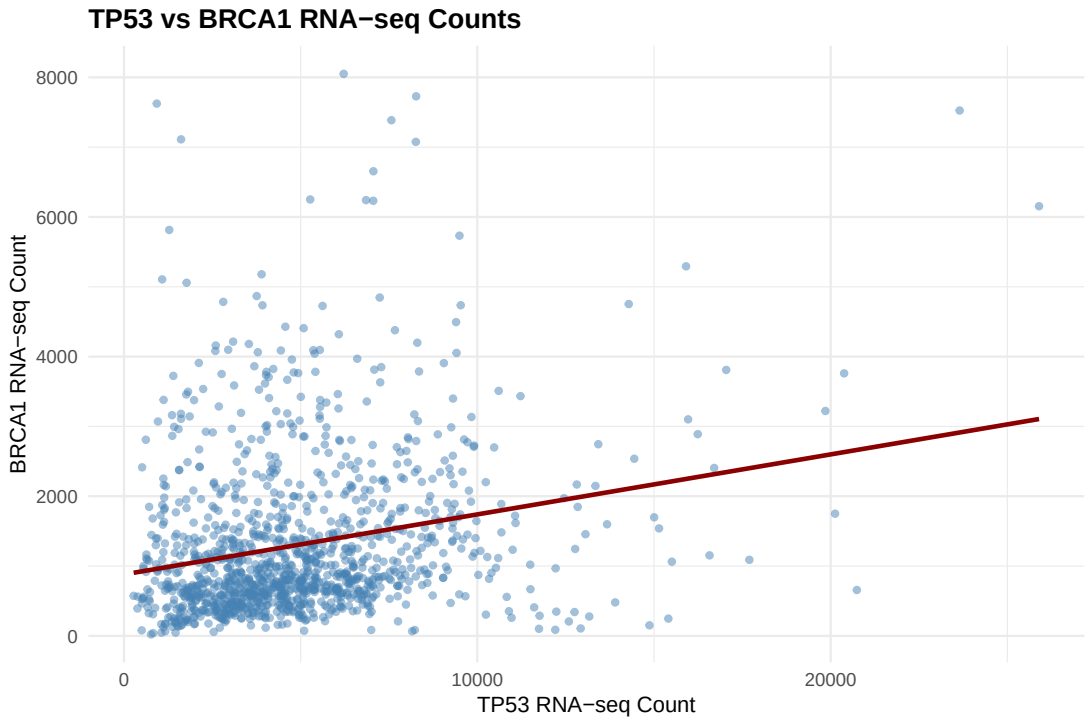


Figure 2: Scatter plot comparing TP53 and BRCA1 RNA-seq counts with a fitted linear trend line.

Figure 2 illustrates the relationship between TP53 and BRCA1 RNA-seq counts.

Pearson's correlation coefficient was $r = 0.2295$, indicating a weak positive linear association between the two genes' raw counts.

The fitted trend line slopes upward, suggesting that samples with higher TP53 counts tend, on average, to have somewhat higher BRCA1 counts. However, the points are widely dispersed around the line, indicating substantial variability and a relatively weak relationship.

The observed correlation does not imply that changes in one gene cause changes in the other. Additionally, shared technical influences on raw counts may contribute to the observed association.

3.5 TP53 Counts Across Sample Types

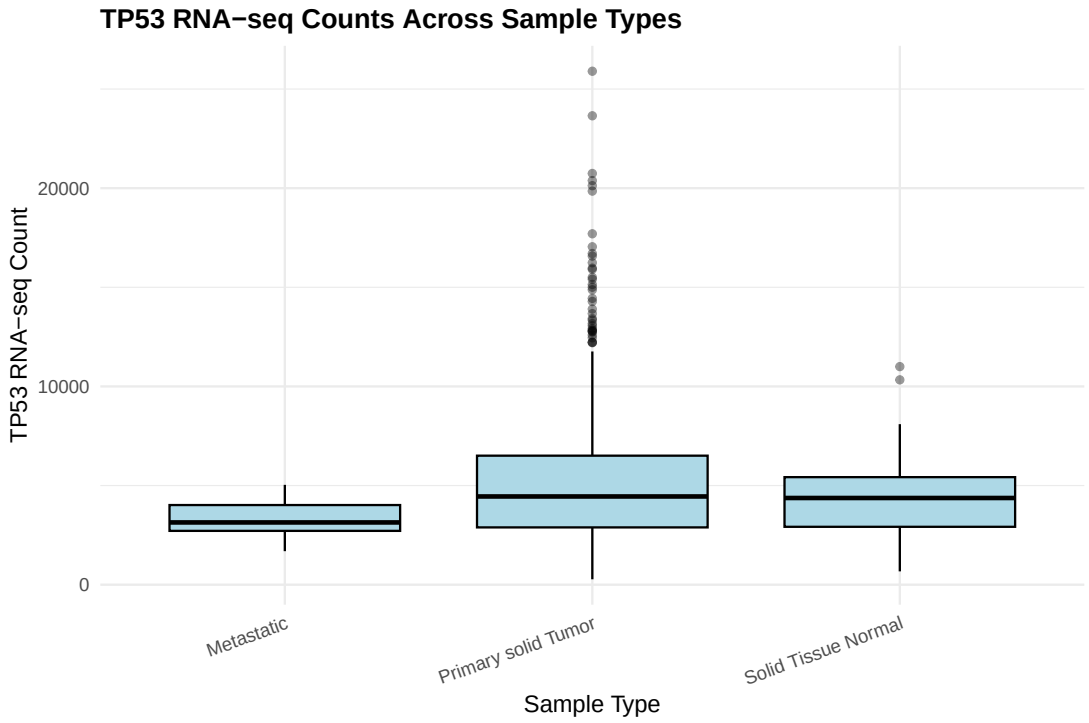


Figure 3: Box plot comparing TP53 RNA-seq counts across metastatic, primary solid tumor, and solid tissue normal samples.

Figure 3 compares the distribution of TP53 counts across the three sample types.

The primary solid tumor category exhibits substantial variability, including numerous observations above the upper whisker of the box plot. Differences in the medians and interquartile ranges are also visible across the sample groups.

These patterns suggest that the distribution of observed TP53 counts is not identical across sample types. However, no formal statistical comparison was performed, and the sample sizes differ substantially.

In particular, the metastatic category contains only seven samples. Therefore, its estimated median and spread should be interpreted with caution.

3.6 Density Distribution of TP53 Counts

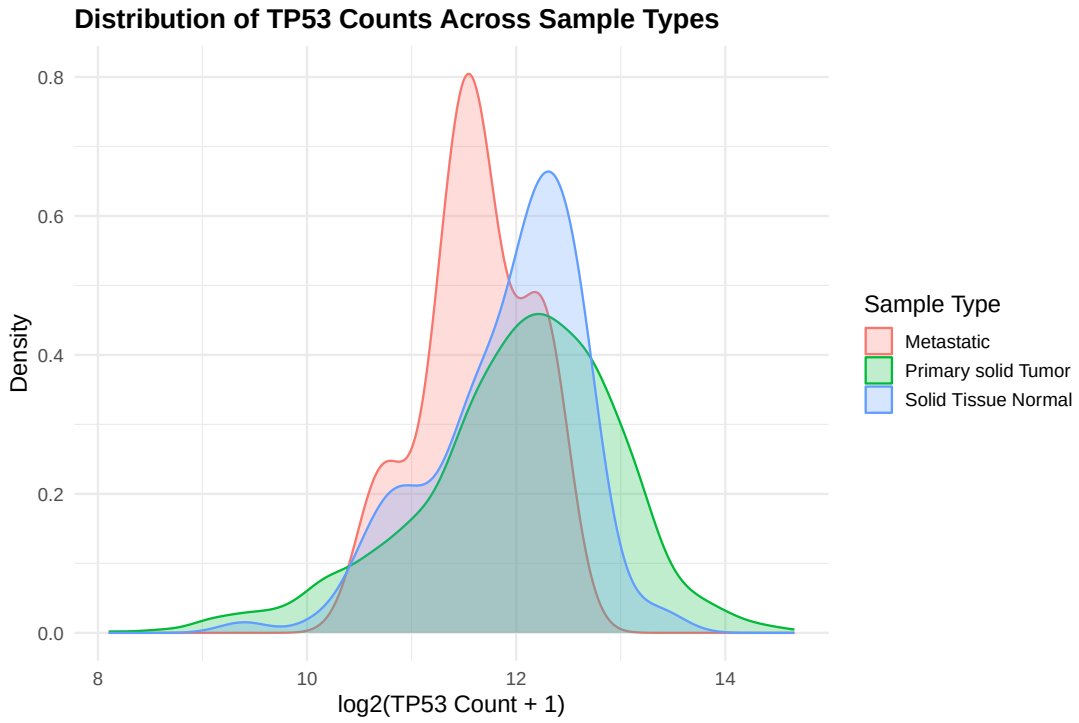


Figure 4: Density distributions of log2-transformed TP53 RNA-seq counts across sample types.

Figure 4 presents an additional visualization of TP53 count distributions by sample type.

The density curves overlap substantially, suggesting that many samples across the categories have similar ranges of transformed TP53 counts.

Nevertheless, differences in the locations of the peaks and the widths of the distributions are visible. The primary solid tumor group has a relatively broad distribution, while the normal tissue group displays a more concentrated peak.

The metastatic density curve should be interpreted particularly cautiously because it was estimated from only seven samples.

The density plot complements the box plot by providing a more detailed view of the overall distribution rather than focusing only on quartiles and outliers.

3.7 Heatmap of Ten Cancer-Related Genes

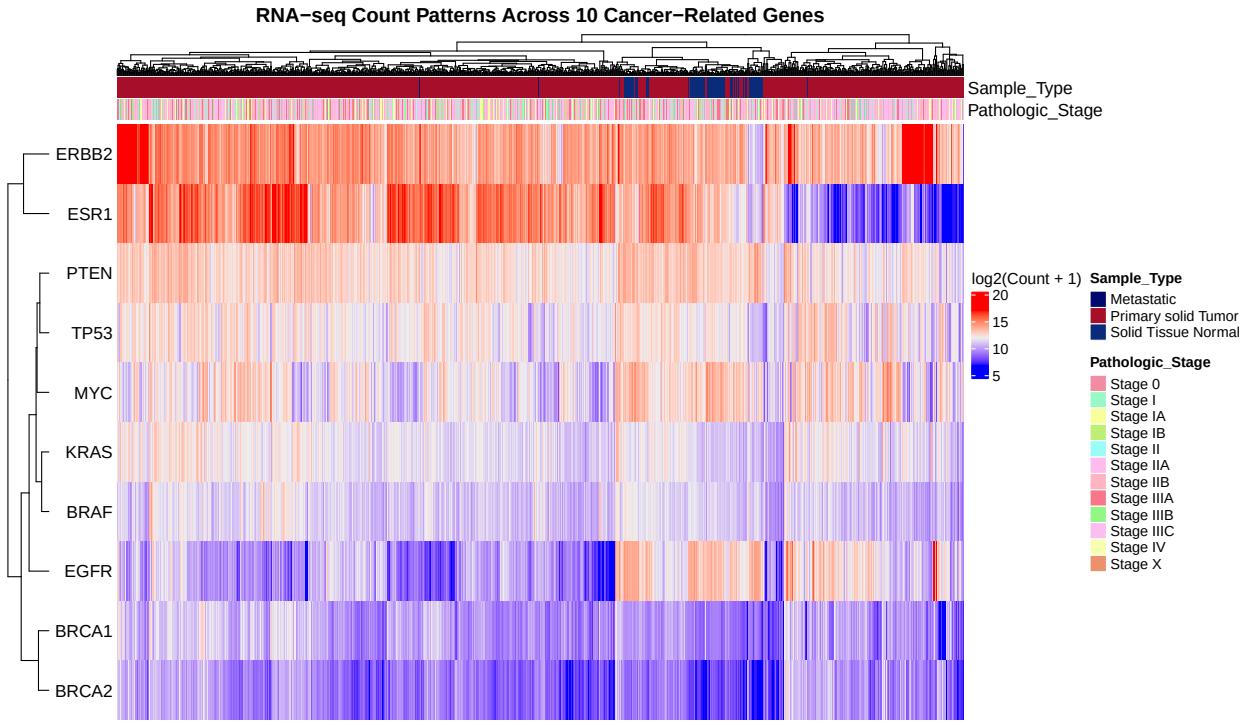


Figure 5: Heatmap of log2-transformed RNA-seq counts for ten cancer-related genes, with sample type and pathological stage annotations.

Figure 5 presents RNA-seq count patterns for ten cancer-related genes across the TCGA-BRCA samples.

The genes included TP53, BRCA1, BRCA2, ERBB2, ESR1, EGFR, PTEN, MYC, KRAS, and BRAF.

A log2 transformation was applied to the counts to improve the visualization of their range. Hierarchical clustering was used to organize genes and samples according to similarities in their count patterns.

The heatmap shows substantial variation across the selected genes and samples. Some genes, including ERBB2 and ESR1, display relatively high transformed counts across many samples, whereas BRCA1 and BRCA2 generally appear at lower values on the displayed scale.

Two categorical annotation bars provide additional information about sample type and pathological stage. These annotations allow the observed count patterns to be considered alongside sample characteristics.

The pathological-stage variable contained 111 missing values, which limits the completeness of this clinical annotation.

Overall, the heatmap demonstrates how multiple genes and clinical variables can be visualized together to explore patterns within a large molecular dataset.

4 Discussion and Limitations

This exploratory analysis identified variation in TP53 and BRCA1 RNA-seq counts across TCGA-BRCA samples. The descriptive statistics and histogram demonstrated substantial right-skewness in the count distributions, while the scatter plot showed a weak positive linear association between TP53 and BRCA1.

The box plot and density plot provided complementary perspectives on TP53 count distributions across sample types. The heatmap extended the analysis by displaying ten cancer-related genes together with sample type and pathological stage annotations.

Several limitations should be considered when interpreting these findings.

First, the analysis used raw RNA-seq counts without normalization for sequencing depth or other technical factors. Raw counts are useful for exploratory visualization, but they are not directly comparable measures of normalized gene expression. More rigorous comparisons between biological groups would require appropriate normalization and statistical methods, such as those described for RNA-seq analysis by Love et al. [3].

Second, the sample categories were highly imbalanced. The dataset contained 1,111 primary solid tumor samples, 113 normal tissue samples, and only seven metastatic samples. This imbalance limits the reliability of comparisons involving the metastatic category.

Third, the pathological-stage metadata contained 111 missing values. Missing clinical information reduces the completeness of the heatmap annotations and may limit interpretation of associations between molecular patterns and clinical characteristics.

Finally, the analysis was descriptive rather than inferential. No differential expression testing, adjustment for confounding variables, or causal analysis was conducted. Accordingly, the observed patterns should be viewed as exploratory findings that could inform subsequent analyses.

Despite these limitations, the project demonstrates how descriptive statistics, multiple visualization types, and categorical annotations can be integrated to explore complex genomic datasets. The analysis also illustrates the importance of clear figure presentation and cautious interpretation when working with high-dimensional biological data.

References

[1] The Cancer Genome Atlas Network. (2012). Comprehensive molecular portraits of human breast tumours. *Nature, 490*, 61–70. https://doi.org/10.1038/nature11412

[2] Roy, R., Chun, J., & Powell, S. N. (2012). BRCA1 and BRCA2: Different roles in a common pathway of genome protection. *Nature Reviews Cancer, 12*, 68–78. https://doi.org/10.1038/nrc3181

[3] Love, M. I., Huber, W., & Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology, 15*, 550. https://doi.org/10.1186/s13059-014-0550-8

[4] Wickham, H. (2016). *ggplot2: Elegant Graphics for Data Analysis*. Springer. https://doi.org/10.1007/978-3-319-24277-4

[5] Gu, Z., Eils, R., & Schlesner, M. (2016). Complex heatmaps reveal patterns and correlations in multidimensional genomic data. *Bioinformatics, 32*(18), 2847–2849. https://doi.org/10.1093/bioinformatics/btw313

[6] National Cancer Institute. Genomic Data Commons Data Portal. https://portal.gdc.cancer.gov/